Video Agent Repository: Multi-Agent Project Experience Summary

Portfolio Narrative for Resume and Interviews

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1 Project Snapshot

- **Project:** Vibe Insta / Video Agent (Python, LangChain, Gemini, YouTube Data API, ElevenLabs, FFmpeg).
- **Objective:** Build an end-to-end AI pipeline that discovers YouTube content opportunities and generates short-form vertical videos.
- **Format target:** 9:16 videos, typically 30–60 seconds, with voiceover, subtitles, and scene visuals.
- **Current maturity:** Functional MVP with documented production-readiness gaps and a prioritized roadmap.

2 Multi-Agent Architecture Implemented

- **MarketResearchAgent** (`src/agent.py`) analyzes long-form vs short-form demand/competition and exports structured `TopicBrief` artifacts.
- **ScriptGenerationAgent** (`src/script_agent.py`) generates fact-grounded scripts with beat-level timing and JSON validation/repair.
- **VideoGenerationAgent** + planner (`src/video_agent.py`, `src/video_planner.py`) convert scripts into deterministic `VideoPlan` scene graphs.
- **AudioGenerationAgent** (`src/audio_agent.py`) produces scene-level TTS segments and `AudioTimeline` manifests, with master-track fallback logic.
- **VisualAssetAgent** (`src/visual_agent.py`) fetches/validates visuals and falls back to deterministic placeholders for offline-safe execution.
- **CompositionAgent** (`src/composition_agent.py`) synchronizes audio + visual layers and emits a renderer-agnostic `RenderSpecification`.
- **RenderAgent** (`src/render_agent.py`) executes FFmpeg or dry-run rendering and outputs final MP4 + metadata.

3 Evidence of Multi-Agent Engineering Skills

- **Contract-first orchestration:** Established explicit artifact handoffs (`TopicBrief` → `ScriptPackage` → `VideoPlan` → `AudioTimeline` → `VisualManifest` → `RenderSpecification` → `FinalVideo`).
- **Determinism and recovery:** Added deterministic ID/path generation, structured validation, and fallback modes (offline script generation, silent audio, placeholder visuals) to keep the pipeline runnable under partial outages.

- **Tool-augmented agents:** Combined LLM reasoning with external tools/services (YouTube API search, transcript extraction with cookie support, TTS generation, FFmpeg rendering) instead of pure prompt chaining.
- **Stateful data grounding:** Integrated SQLite-backed `FactStore` + caption mining/cache so script generation can enforce fact-grounding constraints and avoid unsupported claims.
- **Operational interfaces:** Built modular CLI entry points in `main.py` for stage-wise execution and full pipeline runs (`mvp`, `mvp_offline`) to support debugging and staged deployment.
- **Quality discipline:** Added targeted tests for core pipeline contracts (audio timeline generation, composition synchronization) and maintained roadmap-based gap tracking.

4 Roadmap-Aligned Delivery and Technical Judgment

Completed / Validated

- Implemented text overlay rendering in FFmpeg via `drawtext` and validated on sample outputs.
- Added audio master handoff path (`audio_master` → composition master track → render preference).
- Verified at least one generated MP4 includes a valid AAC audio stream using `ffprobe`.

Known Gaps (Explicitly Tracked)

- Full-duration audio continuity can still mismatch video length in some runs.
- Background music mixing and LUFS normalization remain planned but incomplete.
- Scene-to-image semantic relevance and advanced visual effects are still limited in Phase 1.

5 Resume-Ready Project Bullets (Multi-Agent Focus)

- Designed and implemented a 7-stage multi-agent video generation pipeline with contract-based JSON artifact handoffs from market research through final rendering.
- Engineered deterministic, fault-tolerant orchestration with offline-safe fallbacks (silent TTS, placeholder visuals, dry-run rendering) to preserve end-to-end runnability.
- Built fact-grounded script generation by integrating transcript mining, SQLite fact storage, and schema validation/repair logic for beat-level video scripts.
- Developed an FFmpeg-backed rendering path that composes timed visuals, subtitles, and audio tracks into vertical short-form MP4 outputs.
- Operationalized the system through modular CLI workflows for both stage-level execution and single-command MVP runs.
- Managed delivery using a production-oriented roadmap that distinguishes validated capabilities from known quality and reliability gaps.

6 Keyword Coverage for Applications

- **Core:** Multi-agent systems, agent orchestration, LLM tool use, pipeline design, artifact contracts.
- **ML/LLM:** LangChain, Gemini, prompt design, fact grounding, structured output validation.
- **Systems:** Python, API integration, caching, SQLite, FFmpeg, reliability engineering, fallback design.